

Workings and Solution Notes for Erdős Problem 132

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Problem statement

Let $A \subset \mathbb{R}^2$, $|A| = n$. For each realized distance value $d > 0$, define

$$m_d := \#\{\{a, b\} \subset A : |a - b| = d\}.$$

Define *light distances* by $1 \leq m_d \leq n$.

Questions:

1. Must there exist two distinct light distances?
2. Must the number of light distances tend to infinity with n ?

Stage 1: Reformulation

Set

$$P := \binom{n}{2}, \quad D := \#\{d : m_d > 0\}, \quad L := \#\{d : 1 \leq m_d \leq n\}, \quad H := D - L.$$

Then heavy classes satisfy $m_d > n$, and

$$\sum_d m_d = P.$$

Hence:

- Q1 becomes: is $L \geq 2$?
- Q2 becomes: does $L \rightarrow \infty$ as $n \rightarrow \infty$?

Graph language: each distance value d defines a graph G_d on A , with $|E(G_d)| = m_d$. Light distances are exactly classes with at most n edges.

Stage 2: Immediate inequalities

From $\sum_d m_d = P$:

- Since each heavy class contributes $> n$, one has

$$P > nH \Rightarrow H < \frac{P}{n} = \frac{n-1}{2}.$$

- Therefore

$$L = D - H \geq D - \frac{n-1}{2}.$$

Combining only with Guth–Katz $D \gtrsim n/\log n$ gives

$$L \gtrsim \frac{n}{\log n} - \frac{n}{2},$$

which is asymptotically non-informative for positivity.

Stage 3: Consequences of known bounds

3.1 One unconditional light distance

Let Δ be the diameter of A . The diameter graph in $\mathbb{R}^2$ has at most n edges (Hopf–Pannwitz / Vazsonyi), so

$$m_\Delta \leq n.$$

Hence $L \geq 1$ for every $n \geq 2$.

3.2 Maximum multiplicity does not force many light classes

If $M := \max_d m_d$, then by unit-distance type bounds $M \lesssim n^{4/3}$. Thus

$$D \geq \frac{P}{M} \gtrsim n^{2/3},$$

still too weak to force $L \geq 2$ or $L \rightarrow \infty$.

3.3 Energy bounds remain compatible with few light classes

Let $E_2 := \sum_d m_d^2$. Known upper bounds of order $n^3 \log n$, together with Cauchy lower bound $E_2 \geq P^2/D$, do not exclude spectra where most classes lie near or above n .

Stage 4: Canonical constructions

- Regular n -gon: many classes with multiplicity $\leq n$, so many light distances.
- Square lattice $N \times N$, $n = N^2$: at least $\Omega(N) = \Omega(\sqrt{n})$ light classes from boundary-related displacement vectors.
- Triangular lattice patches: analogous behavior, many light classes.
- Cartesian products / parallel-line constructions: many classes with multiplicity $O(n)$, frequently $\leq n$.
- Circle-heavy constructions: regular/near-regular placements still provide many light chord classes.
- Random sets: almost surely all pair distances distinct, so $L = P = \Theta(n^2)$.

No canonical family produces bounded L .

Stage 5: Contradiction attempt from $L \leq 1$

Assume $L \leq 1$. Then at least $D - 1$ classes are heavy:

$$P = \sum_d m_d \geq 1 + (D - 1)(n + 1),$$

hence

$$D \leq \frac{P - 1}{n + 1} + 1 = \frac{n}{2} + O(1).$$

This is compatible with known lower bounds $D \gtrsim n / \log n$. Energy bounds also remain compatible.

Stage 6: Attempt toward $L \rightarrow \infty$

Dyadic decomposition

$$S_j = \{d : 2^j n < m_d \leq 2^{j+1} n\}$$

combined with incidence machinery controls only very rich classes and yields at best $O(n)$ -scale bounds for heavy classes overall. This still does not force $L \rightarrow \infty$.

Stage 7: Obstruction and partial theorems

- Proven: $L \geq 1$ for all $n \geq 2$ (diameter argument).
- Missing ingredient: strong control on the near-threshold band $m_d \asymp n$, not just high-multiplicity tails.

Stage 8: Final verdict

1. **Literal universal Q1 is false.** For $n = 3$, an equilateral triangle has one distance value d with $m_d = 3 = n$, so $L = 1$, not 2.
2. **Asymptotic questions remain unresolved by current standard tools.** Present bounds do not force either eventual $L \geq 2$ or $L \rightarrow \infty$.

Thus the strongest justified conclusion is:

- always at least one light distance,
- two-light-distance claim fails in finite full generality,
- asymptotic growth of light distances is open under currently available machinery.